CHE-321

Chemical Reaction Engineering

Assignment - 1

Problem1 – 4b

Problem

The Los Angeles basin floor covers approximately 700 square miles $(2 * 10^{10} ft^2)$ and is almost completely surrounded by mountain ranges. If one assumes inversion height in the basin of 2000 feet, the corresponding volume of air in the basin is $4 * 10^{13} ft^3$. Using this system volume we can model the accumulation and depletion of air pollutants. As a very rough first approximation, we shall treat the Los Angeles basin as a well-mixed container (similar to a CSTR) in which there are no spatial variations in pollutant concentrations. We will also simplify the system by considering only the pollutant carbon monoxide and assuming that its only source is from automobile exhaust. On the average we will further suppose that there are 400,000 cars operating in the basin at any one time. Each car gives off approximately 3000 standard cubic feet of exhaust each hour containing 2% mol carbon monoxide.

We shall perform an unsteady-state mole balance on CO as it is depleted from the basin area by a Santa Ana wind. Santa Ana winds are high-velocity winds that originate in the Mojave Desert just to the northeast of Los Angeles. This clean desert air flows into the basin through a corridor assumed to be 20 miles wide and 2000 ft high (inversion height) replacing the polluted air, which flows out to sea or toward the south. The concentration of CO in the Santa Ana wind entering the basin is 0.08 ppm $(2.04 * 10^{-10} \frac{lb*mol}{ft^3})$.

Questions

a. How many pound moles of gas are in the system we have chosen for the Los Angeles basin if the temperature is 75°F and the pressure is 1 atm?

b. What is the rate, $F_{CO,A}$, at which all autos emit carbon monoxide into the basin ($\frac{lb*(moles\ of\ CO)}{h}$)?

c. What is the volumetric flow rate ($\frac{ft^3}{h}$) of a 15 mph wind through the corridor 20 miles wide and 2000 feet high?

d. At what rate, $F_{CO,S}$, does the Santa Ana wind bring carbon monoxide into the basin ($\frac{lb*(moles\ of\ CO)}{h}$)?

e. Assuming that the volumetric flow rates entering and leaving the basin are identical $v = v_0$ show that the unsteady mole balance on CO within the basin becomes $F_{CO,A} + F_{CO,S} - v_O C_{CO} = V\left(\frac{dC_{CO}}{dt}\right)$

f. Verify that the solution to the previous equation is $t = \frac{V}{v_0} ln\left(\frac{[F_{CO,A}+F_{CO,S}]-v_0 C_{CO,O}}{[F_{CO,A}+F_{CO,S}]-v_0 C_{CO,f}}\right)$

g. If the initial concentration of carbon monoxide in the Los Angeles basin before the Santa Ana wind starts to blow is 8 ppm $(2.04 * 10^{-8} \frac{lb\ mol}{ft^3})$, calculate the time required for the carbon monoxide to reach a level of 2 ppm.

h. There is heavier traffic in the L.A. basin in the mornings and in the evenings as workers go to and from work in downtown L.A. Consequently, the flow of CO into the L.A. basin might be better represented by the sine function over a 24-hour period. $F_{CO,A} = a + b * sin(\frac{\pi t}{6})$ The time, t = 0, starts at midnight and $a = 35,000 \frac{lb\ mol}{h}$ and $b = 30,000 \frac{lb\ mol}{h}$. The general material balance equation now becomes $a + b * sin(\frac{\pi t}{6}) + F_{CO,S} - v_O C_{CO} = V\left(\frac{dC_{CO}}{dt}\right)$.

Using Polymath

1. Smog begins to build up again immediately after a Santa Ana wind passes through the basin. The volumetric flow rate through the basin has dropped to $1.67 * 10^{12} \frac{ft^3}{hr}$ (1.5 mph). Plot the concentration of carbon monoxide in the basin as a function of time for up to 72 hours and just after a Santa Ana wind. The initial concentration of CO is 0.08 ppm $(2 * 10^{-10} \frac{lbmol}{ft^3})$.

2. Plot the concentration of carbon monoxide as a function of time t up to 48 hours for the case when a Santa Ana wind begins to blow after the 48 hours buildup of CO.

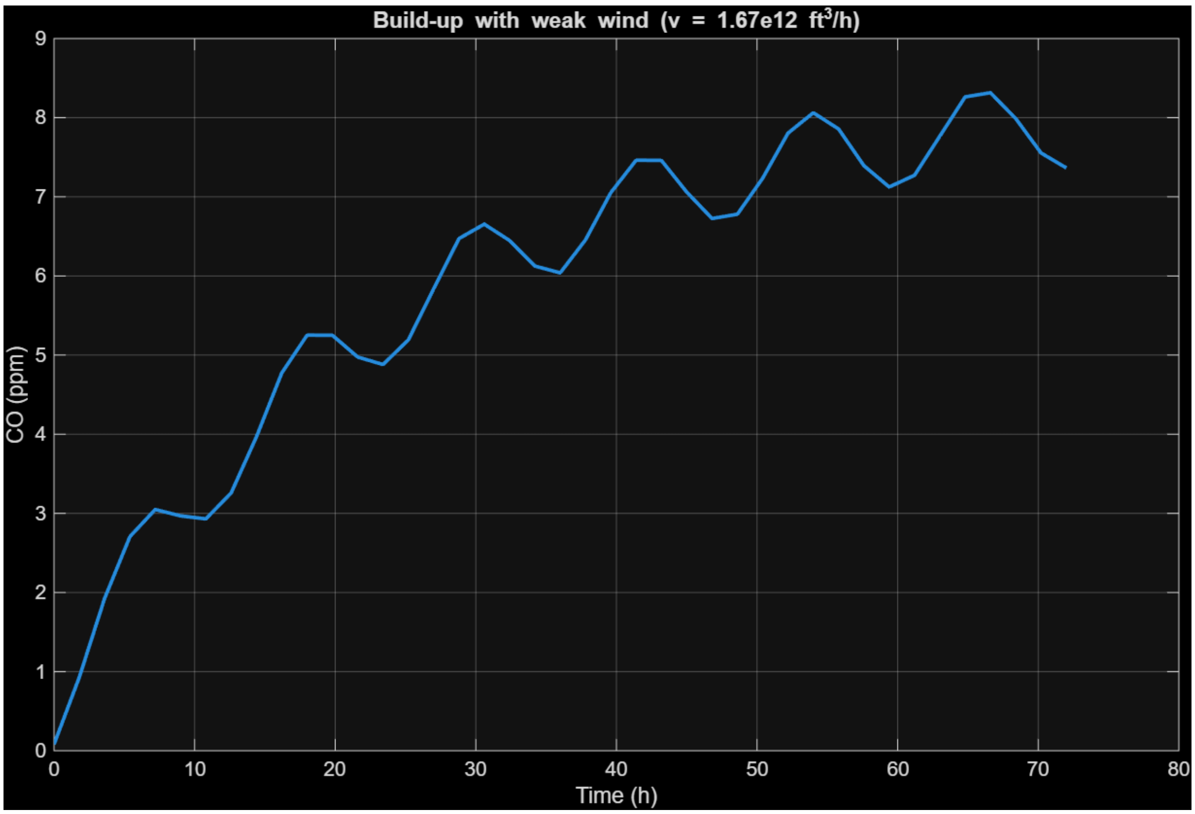
3. Vary the parameters, v_0, a, b, and write a paragraph describing what you find.

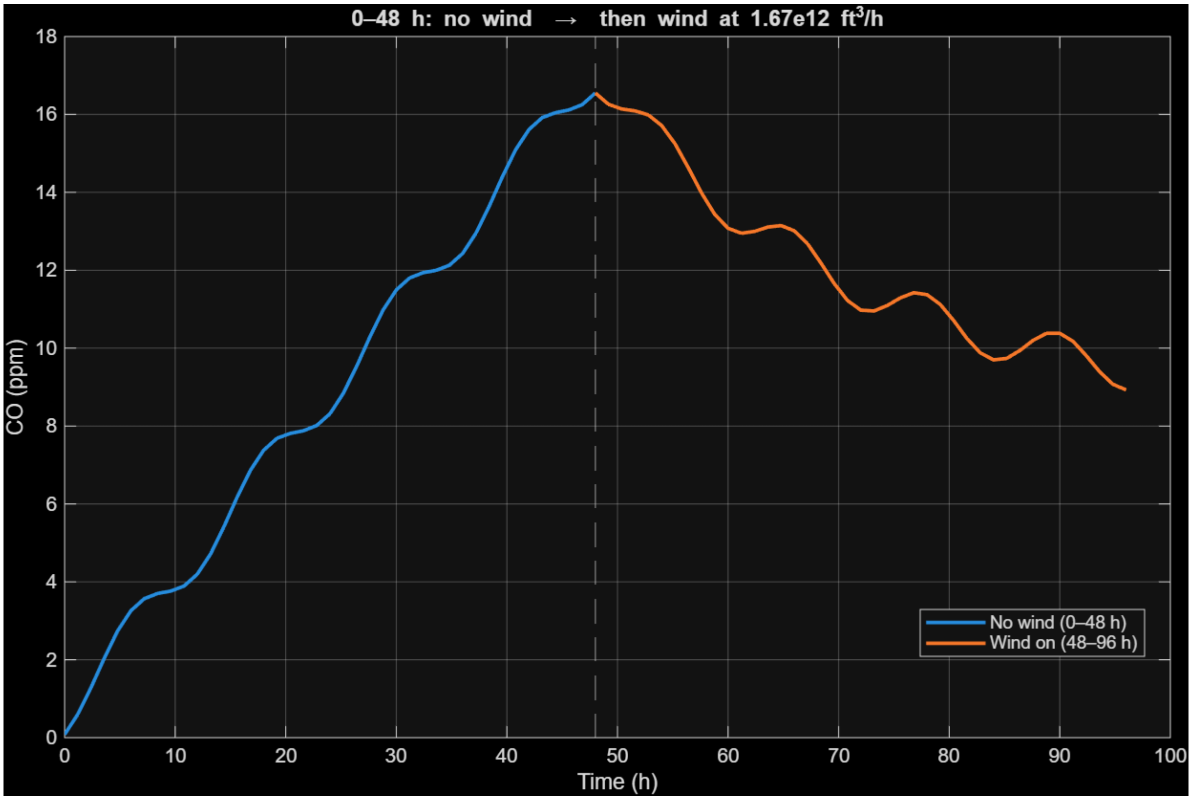
Solutions:

Refer to the matlab code:

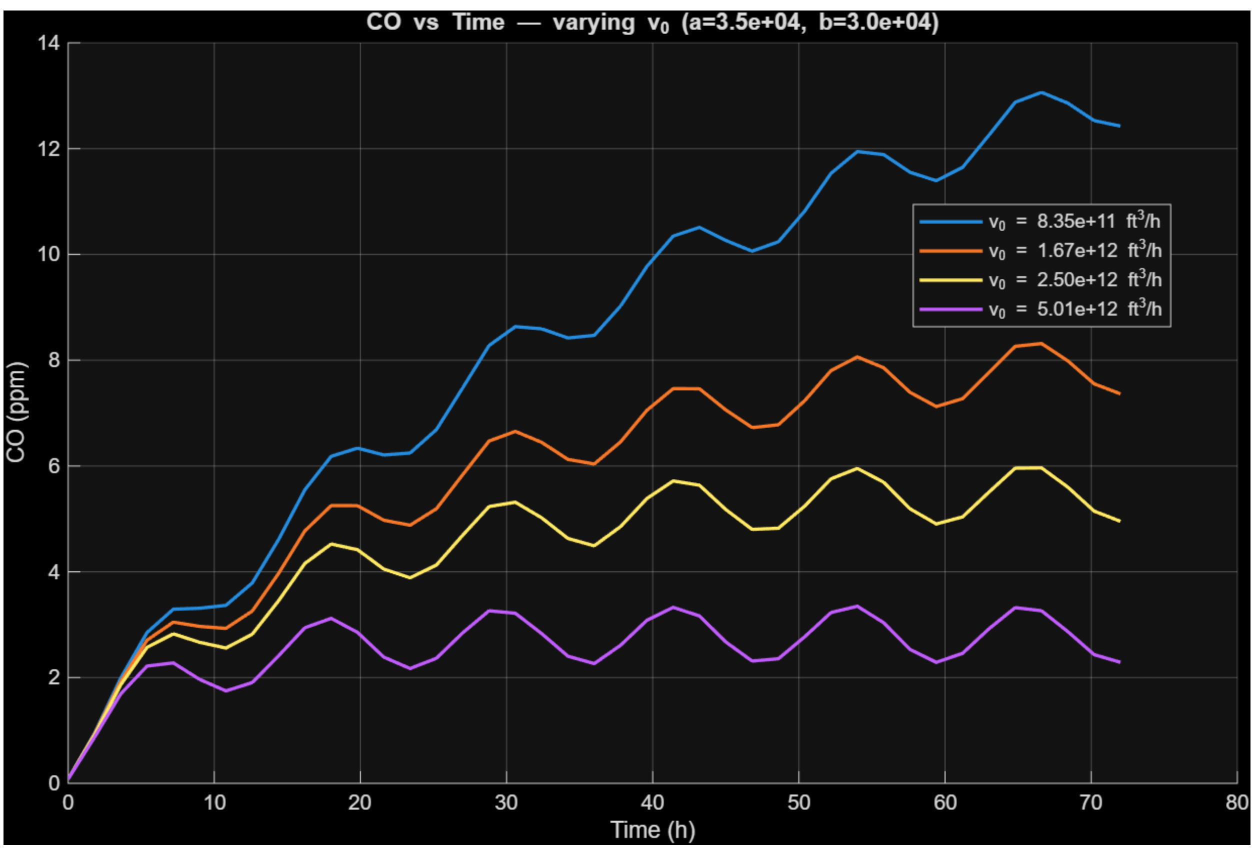
```
Command Window
(a) n_total = 1.025e+11 lbmol
(b) F_CO,A   = 6.685e+04 lbmol/h
(c) v(15mph)= 1.670e+13 ft^3/h
(d) F_CO,S   = 3.407e+03 lbmol/h
(g) t(8 ppm -> 2 ppm) = 6.94 h
```

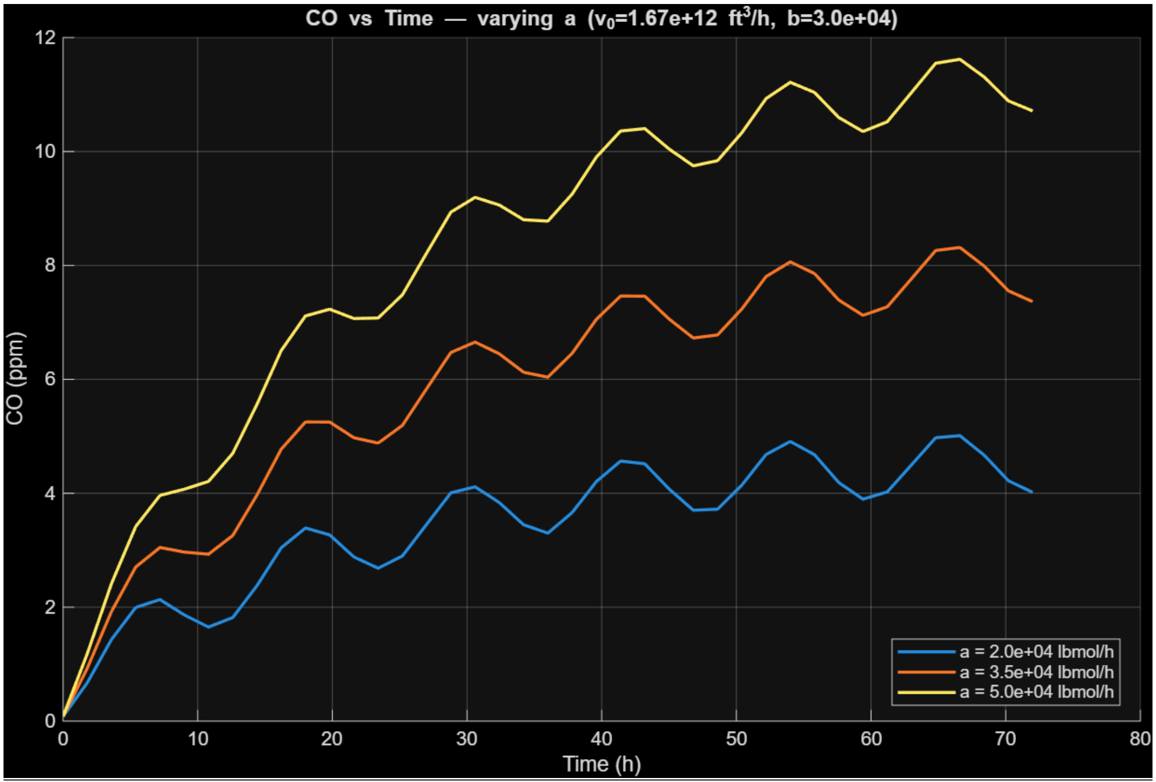
Graphs:

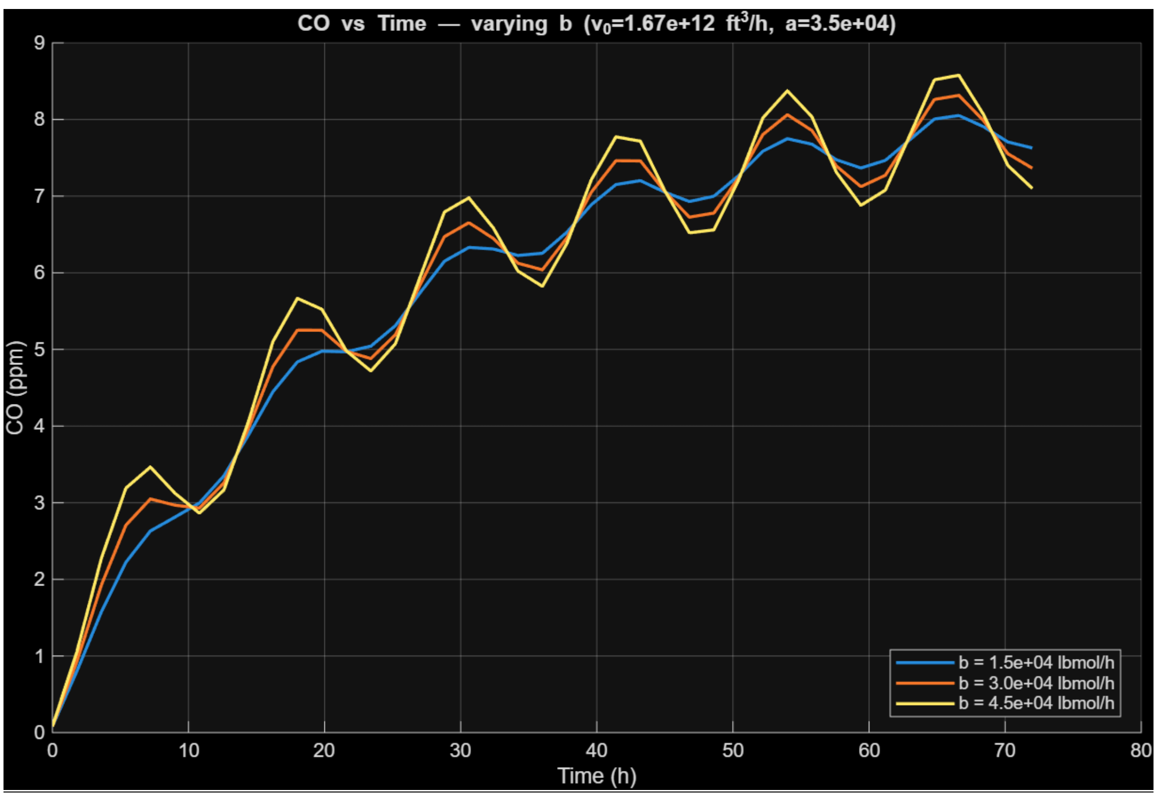




Varying Vo, a and b:




CO vs Time — varying a (v_0=1.67e+12 ft^3/h, b=3.0e+04)
CO (ppm)
Time (h)
a = 2.0e+04 lbmol/h
a = 3.5e+04 lbmol/h
a = 5.0e+04 lbmol/h


CO vs Time — varying b (v_0=1.67e+12 ft^3/h, a=3.5e+04)
CO (ppm)
Time (h)
b = 1.5e+04 lbmol/h
b = 3.0e+04 lbmol/h
b = 4.5e+04 lbmol/h

As Vo increases, both the **average CO level** and the **oscillation amplitude** drop. This matches the first order intuition and the attenuation of a sinusoid by a first-order system. High Vo curves sit lower, are flatter, and track the daily emission waveform more closely; low Vo curves show slow build-up and broader peaks.

As a **raises the entire CO curve** roughly uniformly (mean increases), with only modest changes to amplitude or timing. Over your 72 h window, the change in mean is close to $\Delta\bar{C} \approx \Delta a/v_0$. With weak wind (baseline Vo), roughly doubles the offset above background.

Increasing bbb **widens the peak–trough spread** but leaves the long-term mean nearly unchanged (mean governed by a, not b). The amplitude gain is not 1:1—stronger ventilation damps the oscillation.

<u>Matlab code:</u>

```
clear; clc; close all;


V   = 4.0e13;

C_in = 2.04e-10;

ppm2c = @(ppm) ppm*(2.04e-10/0.08);

c2ppm = @(c)  c /(2.04e-10/0.08);




nCars   = 4.0e5;

exh_scf = 3000;

Tstd_R  = 32 + 459.67;

scf_per_lbmol = 0.7302*Tstd_R;

%b

F_CO_A_const = 0.02*(nCars*exh_scf)/scf_per_lbmol;
```

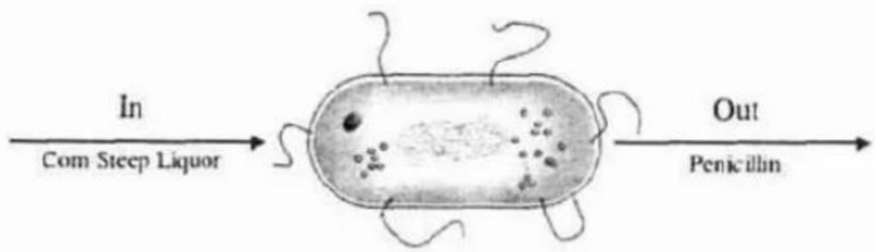
```
%c
v_wind_hi = 1.67e13;

v_wind_lo = 1.67e12;

a = 3.5e4;  b = 3.0e4;
ode = @(t,C,v) (a + b*sin(pi*t/6) + v*C_in - v*C)/V;

%a
R = 0.7302; T_R = 75+459.67;
n_total = (1*V)/(R*T_R);
fprintf('(a) n_total = %.3e lbmol\n', n_total);
fprintf('(b) F_CO,A  = %.3e lbmol/h\n', F_CO_A_const);
fprintf('(c) v(15mph)= %.3e ft^3/h\n', v_wind_hi);
%d
fprintf('(d) F_CO,S  = %.3e lbmol/h\n', v_wind_hi*C_in);

v_cars = nCars*exh_scf;
nu0 = v_wind_hi + v_cars;
C0 = ppm2c(8);  Cf = ppm2c(2);
num = F_CO_A_const + v_wind_hi*C_in - nu0*C0;
den = F_CO_A_const + v_wind_hi*C_in - nu0*Cf;
t_g = (V/nu0)*log(num/den);
```

```
fprintf('(g) t(8 ppm -> 2 ppm) = %.2f h\n', t_g);



tspan1 = [0 72];
C0_clean = ppm2c(0.08);
[t1,C1] = ode45(@(t,C) ode(t,C,v_wind_lo), tspan1, C0_clean);

figure; plot(t1, c2ppm(C1), 'LineWidth', 1.8);
xlabel('Time (h)'); ylabel('CO (ppm)'); grid on;
title('Build-up with weak wind (v = 1.67e12 ft^3/h)');

[t_off, C_off] = ode45(@(t,C) ode(t,C,0.0), [0 48], C0_clean);
C_init_on = C_off(end);
[t_on,  C_on]  = ode45(@(t,C) ode(t,C,v_wind_lo), [0 48], C_init_on);

figure;
plot(t_off, c2ppm(C_off), 'LineWidth', 1.8); hold on;
plot(48+t_on, c2ppm(C_on), 'LineWidth', 1.8);
xline(48,'--'); grid on; xlabel('Time (h)'); ylabel('CO (ppm)');
title('0–48 h: no wind → then wind at 1.67e12 ft^3/h');
legend('No wind (0–48 h)','Wind on (48–96 h)','Location','best');
```

Changing

Problem 1 – 11b

P1-11$_B$ We are going to consider the cell as a reactor. The nutrient corn steep liquor enters the cell of the microorganism *Penicillium chrysogenum* and is decomposed to form such products as amino acids, RNA, and DNA. Write an unsteady mass balance on (a) the corn steep liquor, (b) RNA, and (c) penicillin. Assume the cell is well mixed and that RNA remains inside the cell.



Penicillium chrysogenum

We know that,

IN - OUT + GENERATION - CONSUMPTION = ACCUMULATION

(a) Corn Steep liquor

It enters & is consumed

NO intrinsic generation

$$V \frac{dC_s}{dt} = F_{in} C_{s,in} - F_{out} C_s - (-r_s) V$$

Since 's' is only consumed

we can write

$$\boxed{V \frac{dC_s}{dt} = F_{in} C_{s,in} - F_{out} C_s - R_{s,cons} V}$$

$\boxed{r_s < 0}$

(b) RNA (R)

RNA is made inside & remains inside the cell

Hence,

$$\boxed{\frac{V\,dC_s}{dt} = r_R V}$$

(c) Penicillin (P)

Formed inside the cell & exported; none in the feed.

Hence,

$$\boxed{\frac{V\,dC_p}{dt} = 0 - F_{out} C_p - r_p + V}$$